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Title: Homomorphisms of partial t-trees and edge-coloring of partial 3-trees

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Abstract: A reformulation of the four-color theorem is to say that K4 is the smallest graph to which every planar (loop-free) graph admits a homomorphism. Extending this theorem, the second author has proved (using the four-color theorem) that the Clebsch graph (a well known triangle-free graph on 16 vertices) is a smallest graph to which every triangle-free planar graph admits a homomorphism. As a further generalization he has proposed that the projective cube of dimension 2k, PC(2k), (that is the Cayley graph (Z_2 ^{2k}, {e1, e2, ... e_2k, J}), where ei's are the standard basis and J = e1+e2+...+ e2k) is a smallest graph of odd-girth 2k + 1 to which every planar graph of odd-girth at least 2k + 1 admits a homomorphism. This conjecture is related to a conjecture of P. Seymour who claims that the fractional edge-chromatic number of a planar multigraph determines its edge-chromatic number (more precisely, Seymour conjectures that X'(G) = \lceil X'_f(G)\rceil for any planar multigraph G). Noting that the restriction of Seymour's conjecture on cubic graphs is Tait's reformulation of the four-color theorem. Both these conjectures are believed to be true for the larger class of K5-minor-free graphs (which includes the class of planar graphs). Motivated by these conjectures and in extension of recent work of L. Beaudou, F. Foucaud and the second author, which studies homomorphism bounds for the class of K4-minor-free graphs, in this work we first give a necessary and sufficient condition for a graph B of odd-girth 2k+1 to admit a homomorphism from any partial t-tree of odd-girth at least 2k+1. Applying our results on the class of partial 3-trees, which is a rich subclass of K5-minor-free graphs, we prove that PC(2k) is in fact a smallest graph of odd-girth 2k + 1 to which every partial 3-tree of odd-girth at least 2k + 1 admits a homomorphism. We then apply this result to show that every planar (2k + 1)-regular multigraph G whose dual is a partial 3-tree, and whose fractional edge-chromatic number is 2k + 1, is (2k+1)-edge-colorable. Both these results are the best known supports for the general cases of the above mentioned conjectures in extension of the four-color theorem.

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Homomorphisms of partial t-trees and edge-coloring of partial 3-trees

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Abstract

A reformulation of the four-color theorem is to say that K_4 is the smallest graph to which every planar (loop-free) graph admits a homomorphism. Extending this theorem, the second author has proved (using the four-color theorem) that the Clebsch graph (a well known triangle-free graph on 16 vertices) is a smallest graph to which every triangle-free planar graph admits a homomorphism. As a further generalization he has proposed that the projective cube of dimension $2k$, $PC(2k)$, (that is the Cayley graph $(\mathbb{Z}_2^{2k}, \{e_1, e_2, \ldots, e_{2k}, J\}$, where e_i's are the standard basis and $J = e_1 + e_2 + \cdots + e_{2k})$ is a smallest graph of odd-girth $2k+1$ to which every planar graph of odd-girth at least $2k+1$ admits a homomorphism. This conjecture is related to a conjecture of P. Seymour who claims that the fractional edge-chromatic number of a planar multigraph determines its edge-chromatic number (more precisely, Seymour conjectures that $\chi'(G) = \lceil \chi'_f(G) \rceil$ for any planar multigraph G). Noting that the restriction of Seymour's conjecture on cubic graphs is Tait's reformulation of the four-color theorem.

Both these conjectures are believed to be true for the larger class of K_5-minor-free graphs (which includes the class of planar graphs). Motivated by these conjectures and in extension of recent work of L. Beaudou, F. Foucaud and the second author, which studies homomorphism bounds for the class of K_4-minor-free graphs, in this work we first give a necessary and sufficient condition for a graph B of odd-girth $2k+1$ to admit a homomorphism from any partial t-tree of odd-girth at least $2k+1$. Applying our results on the class of partial 3-trees, which is a rich subclass of K_5-minor-free graphs, we prove that $PC(2k)$ is in fact a smallest graph of odd-girth $2k+1$ to which every partial 3-tree of odd-girth at least $2k+1$ admits a homomorphism. We then apply this result to show that every planar $(2k+1)$-regular multigraph G whose dual is a partial 3-tree, and whose fractional edge-chromatic number is $2k+1$, is $(2k+1)$-edge-colorable. Both these results are the best known supports for the general cases of the above mentioned conjectures in extension of the four-color theorem.

Key words: Planar graphs, treewidth, homomorphisms, minor, edge-coloring.

1. Introduction

Given graphs G and H, a *homomorphism* of G to H is a mapping of the vertices of G to the vertices of H which preserves adjacency, that is to say, a mapping $h : V(G) \to V(H)$ such that if x and y are adjacent in G then $h(x)$ and $h(y)$ are adjacent in H. When there exists a homomorphism of G to H we write $G \to H$ and we may say G *maps* to H. It is easily verified that mapping of G to K_k corresponds to a k-coloring of G, thus homomorphism generalizes the theory of coloring of graphs. Given a class $\mathcal{C}$ of graphs we say H *bounds* $\mathcal{C}$ if every member of $\mathcal{C}$ maps to H. Thus, in this terminology, the four-color theorem is to say: K_4 bounds the class of planar graphs.

Given a graph G, the *odd-girth* of G is the length of a shortest odd-cycle of G. It can be easily verified that an odd-cycle of length $2l+1$ maps to an odd-cycle of length $2k+1$ if and only if $l \geq k$. Thus we have the following no-homomorphism lemma:

Lemma 1.1. *If $G \to H$, then odd-girth$(G) \geq$ odd-girth(H).*

In fact slightly stronger statement can be made:

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Lemma 1.2. *Let G and B be graphs of odd-girth $2k+1$ and suppose that there is a homomorphism ϕ of G to B. Let u and v be two vertices of G which belong to a common $(2k+1)$-cycle of G. Then $\phi(u)$ and $\phi(v)$ are on a common $(2k+1)$-cycle of B and $d_G(u,v) = d_B(\phi(u), \phi(v))$.*

Observe that odd-girth of a given graph G can be computed in time polynomial of the order of G. Given a vertex x of G compute, inductively, the set $N_i(x)$ consisting of vertices at distance i from x. The first i for which $N_i(x)$ induces an edge implies existence of a closed-walk of length $2i+1$ starting at x. The smallest of such closed walks is a shortest odd-cycle of G. Thus this classic no-homomorphism lemma is among rare no-homomorphism lemmas where a proof of a no-homomorphism is based on a parameter computable in polynomial time (other well-known no-homomorphism lemmas are based on parameters such as the clique number, the chromatic number, the fractional and circular chromatic numbers, all of which are NP-hard to compute).

The simplest use of this lemma concerns loops that are odd-cycles of length 1, it thus claims when H is a simple graph and $G \to H$, then G must have no loops. Hence, in this work we consider only graphs with no loops. However multi-edges will be considered when dealing with application on edge-coloring. When multi-edges are allowed to exist we will clarify using the term *multigraph*, thus making sure the term graph refers solely to a simple graph.

In view of this no-homomorphism lemma the four-color theorem can be restated as follows:

Theorem 1.3 (4CT)**.** *The smallest graph with no loop which bounds the class of all planar graphs with no loop is K_4.*

The natural question then is to find a (or probably the unique) smallest graph of odd-girth at least $2k+1$ which bounds the class of all planar graphs of odd-girth $2k+1$. A conjecture of the second author which implies, in particular, the order of such a graph must be 2^{2k} is discussed in Section 5.

Motivated by this question and the recent work of [2] we consider the question of bounding the class of partial t-trees of odd-girth at least $2k+1$ by a graph of odd-girth $2k+1$. A necessary and sufficient condition for a graph B of odd-girth $2k+1$ to admit a homomorphism from all K_4-minor-free graphs of odd-girth at least $2k+1$ is provided in [2]. Extending this work we give a necessary and sufficient condition to test if a given graph B of odd-girth $2k+1$ bounds the class of partial t-trees of odd-girth at least $2k+1$. We will then apply our work to the class of partial 3-trees to obtain results which are the strongest known support for the general cases of some conjectures in generalization of the four-color theorem.

The paper is organized as follows. In the next section we pick our adaptation of various terminology for the class of partial t-trees. In the section following this, extending the notion of odd-girth to weighted graphs we build up the required terminology and provide some classification. Then in Section 4 we prove our main theorem. In Section 5 we discuss a conjecture in generalization of the four-color theorem, then we provide support for the conjecture by finding an optimal bound of odd-girth $2k+1$ for partial 3-trees of odd-girth at least $2k+1$. In Section 6 we consider an edge-coloring conjecture of P. Seymour on the edge-chromatic number of planar graphs which also extends the four-color theorem. We provide support for this conjecture by proving it for a subclass of multigraphs which are planar and whose duals are partial 3-trees.

2. Partial t-trees

A t-tree is any graph which can be built by the following procedure.

- K_t is a t-tree.
- Given a t-tree T and a t-clique C of T, the graph T' built from T by adding a vertex which is adjacent to all the vertices of C is also a t-tree.

After an arbitrary ordering of the vertices of the first K_t, this construction introduces an ordering $v_1, v_2, \ldots, v_n$ of the vertices which has the following properties: *i.* the subgraph induced by $v_1, v_2, \ldots, v_t$ is a clique, *ii.* in the subgraph H_i induced by vertices $v_1, v_2, \ldots, v_i$, $i > t$, the vertex v_i is of degree t and its neighbors induces a complete graph on t-vertices. Such an ordering of the vertices of a t-tree T is called a *t-tree ordering* of T. Given a t-tree T and a t-tree ordering $v_1, v_2, \ldots, v_n$, $n > t$, and for $t+1 \leq i \leq n$, let X_i be the $(t+1)$-clique induced by v_i and all its nighbour among $\{v_1, v_2, \ldots, v_{i-1}\}$.

The sequence $X_{t+1}, X_{t+2}, \ldots, X_n$ which is uniquely determined by the given t-tree ordering of vertices is called a *clique-sequence* of T.

Observe that isomorphic copies of a t-tree might be built by different t-tree orderings. Up to such isomorphisms, the class of 1-trees is exactly the class of trees, granting the name t-tree.

A *partial t-tree* is any subgraph of a t-tree. A partial t-tree G might be a subgraph of two or more non-isomorphic t-trees on a same set $V(G)$ of vertices. Given a partial t-tree G, any t-tree ordering of a t-tree T, which is built on $V(G)$ and contains G as a subgraph, is also a t-tree ordering of G. We will denote the class of all partial t-trees by $\mathcal{PT}_t$ and the subclass of partial t-trees of odd-girth at least $2k+1$ by $\mathcal{PT}_{t,2k+1}$.

The class $\mathcal{PT}_t$ is known under various equivalent definitions. It is most notably known as the class of graphs of treewidth at most t. It is easily verified that $\mathcal{PT}_t$ is a minor-closed class of graphs. For $t = 1$ we have the class of all forests which is identified as the class of graphs with no K_3-minor. The class $\mathcal{PT}_2$ is the class of K_4-minor-free graphs, for a proof see for example [6]. The next class, $\mathcal{PT}_3$, is the class of all graphs having non of the four graphs of Figure 1 as minor, proved in [1]. For $t \geq 4$ the full list of forbidden minors are not known, though they are finite thanks to Graph-Minor theorem of Rebertson-Seymour. One thing sure though: graphs in $\mathcal{PT}_t$ are K_{t+2}-minor-free but K_{t+1} is a member of this class. In particular $\mathcal{PT}_3$ forms a special subclass of K_5-minor-free graphs.

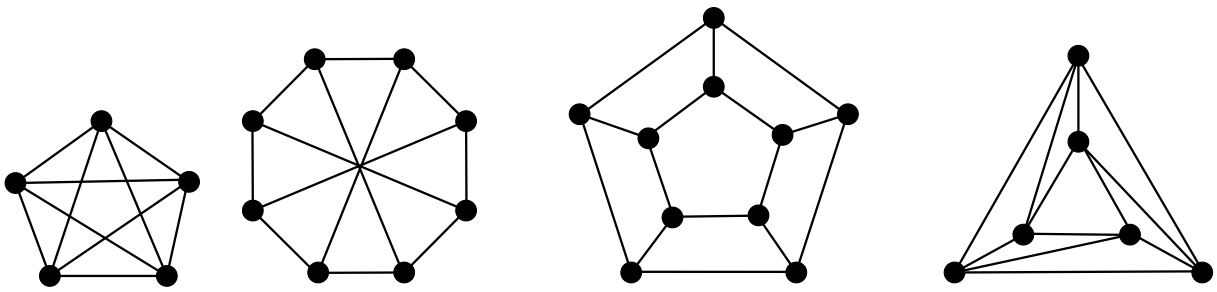

Figure 1: Forbidden minors of the class of partial 3-trees

3. Weighted graphs and notion of $(2k+1)$-wide

While previous notation were repeat of classic terminology, here we introduce some notion we have developed to address our work. Throughout this subsection and also when it is clear from the context, k is a given positive integer where $2k+1$ would be a lower bound on the odd-girth of graphs and weighted graphs we are working with, notion of odd-girth for weighted graphs being defined below.

Given a graph G the multigraph built from G by adding a parallel edge e' for each edge e of G is denoted by $2G$.

A *weighted graph* (G, w) is a graph together with an assignment $w : E(G) \to \mathbb{Z}^+$ which assigns a positive integer $w(e)$ to each edge e of G.

A homomorphism of a weighted graph (G, w_1) to a weighted graph (H, w_2) is a homomorphism of G to H which preserves the weights.

Given a weighted graph (G, w) satisfying $w(e) \leq 2k$, we define $\overline{\overline{(G,w)}}_{2k+1}$ to be a graph built from $2G$ by replacing each edge e with a path of length $w(e)$ and its parallel edge e' with a path of length $2k+1-w(e)$ (all internal vertices of these paths being distinct). Recall that a path in a graph is called *thread* if all its internal vertices are of degree 2. Thus $\overline{\overline{(G,w)}}_{2k+1}$ can be seen as a graph built over $V(G)$ where for each edge xy of G two threads (whose internal vertices are new) connect x and y, one of these threads being of length $w(xy)$, the other being of length $2k+1-w(xy)$. For an example see Figure 2.

We then say (G, w) is $(2k+1)$-*wide* if the graph $\overline{\overline{(G,w)}}_{2k+1}$ is of odd-girth $2k+1$. Thus whenever we claim (G, w) is $(2k+1)$-wide we implicitly imply that w satisfies $1 \leq w(e) \leq 2k$ for all edges of G.

Given a weighted graph (G, w) where w satisfies $w(e) \leq 2k$ for all edges of G, if a new weight w' is obtained by changing weight of an edge e with $w'(e) = 2k+1-w(e)$, then $\overline{\overline{(G,w')}}_{2k+1}$ is isomorphic to $\overline{\overline{(G,w)}}_{2k+1}$. Thus we may freely assume weights of all edges are bounded above by k. However, when

concerned with the parity of a cycle in $\overline{\overline{(G,w)}}_{2k+1}$ going through three or more original vertices of G we may need to consider some values of w which are greater than k.

As a corollary of Lemma 1.2, we have the following lemmas:

Lemma 3.1. *Given two weighted graphs (G,w_1) and (H,w_2) where w_1 and w_2 are both bounded above by $2k$ if $(G,w_1) \to (H,w_2)$ and (H,w_2) is $(2k+1)$-wide, then so is (G,w_1).*

Lemma 3.2. *Given two weighted graphs (G,w_1) and (H,w_2) which are both $(2k+1)$-wide, we have $(G,w_1) \to (H,w_2)$ if and only if we have a homomorphism of the graph $\overline{\overline{(G,w_1)}}_{2k+1}$ to the graph $\overline{\overline{(H,w_2)}}_{2k+1}$.*

Next we would like to have a test which would help to decide whether a given weighted complete graph is $(2k+1)$-wide. We first give two independent tests to do this job for weighted triangles. We then show that for weighted complete graphs of larger order it is enough to apply the test on all induced triangles. The first test applies when we assume all weights are bounded above by k, this test is proved in [2].

Proposition 3.3. *[2] Let (K_3,w) be a weighted triangle of weights a,b,c satisfying $1 \le a \le b \le c \le k$. Then it is $(2k+1)$-wide if and only if one of the following holds:*
(i) $a+b+c$ is odd and $a+b+c \ge 2k+1$,
(ii) $a+b+c$ is even and $a+b \ge c$.

The next test is when w is allowed to give also values between k and $2k$. We prove this test here.

Proposition 3.4. *Let (K_3,w) be a weighted triangle of weights a,b,c satisfying $1 \le a,b,c \le 2k$ such that $a+b+c$ is odd. Let $f_{2k+1}(a,b,c) = \frac{1}{2}(a+b+c-(2k+1))$. Then (K_3,w) is $(2k+1)$-wide if and only if $0 \le f_{2k+1}(a,b,c) \le \min\{a,b,c\}$.*

Proof. Suppose (K_3,w) is $(2k+1)$-wide. By the definition, the graph $\overline{\overline{(K_3,w)}}_{2k+1}$ is of odd-girth $2k+1$. Then each odd-cycle of $\overline{\overline{(K_3,w)}}_{2k+1}$ has length at least $2k+1$. Let x,y,z be three vertices of K_3. There are eight cycles in $\overline{\overline{(K_3,w)}}_{2k+1}$ containing all three vertices x,y and z. Exactly four of these eight cycles are of odd length, their corresponding length are $a+b+c$, $(2k+1-a)+(2k+1-b)+c$, $(2k+1-a)+b+(2k+1-c)$, $a+(2k+1-b)+(2k+1-c)$. Since there is an odd-cycle of length $a+b+c$, and since we have assumed all odd-cycles are of length at least $2k+1$, we get $a+b+c \ge 2k+1$, thus $f_{2k+1}(a,b,c) \ge 0$. The odd-cycle corresponding to the length $2k+1-a+2k+1-b+c$ implies $2k+1-a+2k+1-b+c \ge 2k+1$, therefore $f_{2k+1}(a,b,c) \le c$. Similarly, we have $f_{2k+1}(a,b,c) \le b$ and $f_{2k+1}(a,b,c) \le a$. So $0 \le f_{2k+1}(a,b,c) \le \min\{a,b,c\}$.

Conversely, assume $0 \le f_{2k+1}(a,b,c) \le \min\{a,b,c\}$, we want to show that (K_3,w) is $(2k+1)$-wide. By the definition we need to show that the graph $\overline{\overline{(K_3,w)}}_{2k+1}$ is of odd-girth $2k+1$. The odd-cycles in $\overline{\overline{(K_3,w)}}_{2k+1}$ which contain exactly two vertices of K_3 are of length $2k+1$. There are four odd-cycles which contain three vertices of K_3. Since $a+b+c$ is odd, these four odd-cycles are of length $a+b+c$, $2k+1-a+2k+1-b+c$, $2k+1-a+b+2k+1-c$, $a+2k+1-b+2k+1-c$. By the assumption that $f_{2k+1}(a,b,c) \ge 0$, we get $a+b+c \ge 2k+1$. Since $f_{2k+1}(a,b,c) \le c$, we have $2k+1-a+2k+1-b+c \ge 2k+1$. Similarly, we get that the other two odd-cycles have length at least $2k+1$. This completes the proof. □

Next we prove that: given a weighted complete graph (K_t,w) to decide whether it is $(2k+1)$-wide, it would enough to apply either one of the previous two propositions on all induced triangles.

Theorem 3.5. *A complete weighted graph (K_t,w) is $(2k+1)$-wide if and only if each of its induced triangles is $(2k+1)$-wide.*

Proof. First we show that the condition is necessary. Assume that (K_t,w) is $(2k+1)$-wide. Then, by the definition, $1 \le w(e) \le 2k$ for any edge $e \in E(K_t)$ and the graph $\overline{\overline{(K_t,w)}}_{2k+1}$ is of odd-girth $2k+1$. Let (K_3^*,w^*) be an induced triangle where w^* is induced by w over the edges of K_3^*. Thus $\overline{\overline{(K_3^*,w^*)}}_{2k+1}$, is a subgraph of $\overline{\overline{(K_t,w)}}_{2k+1}$ and, therefore, is also of odd-girth $2k+1$.

To prove that the condition is also sufficient, we assume that (K_t,w) is not $(2k+1)$-wide and we show that there exists a set of three vertices whose induced weighted triangle is not $(2k+1)$-wide. Since

(K_t, w) is not $(2k+1)$-wide, there exists an odd-cycle in $\overline{\overline{(K_t, w)}}_{2k+1}$ of length less than $2k+1$. Assume C is such an odd-cycle with minimum number of vertices from K_t. By the construction C must have at least three vertices of K_t. If exactly three, then we have found our triangle. Thus we assume that C has at least four vertices from K_t. Let x, y, z, u be four of such vertices in clockwise direction of C. Denote the two paths in C connecting x and z by P and Q, we know the length of P and Q have different parity. There are two threads in $\overline{\overline{(K_t, w)}}_{2k+1}$ which connect x and z, denote them by R_1 and R_2. The sum of the length of R_1 and R_2 is $2k+1$, so the length of R_1 and R_2 have also different parity. Without loss of generality, assume that $P \cup R_1$ and $Q \cup R_2$ are odd-cycles (if necessary, we relabel R_1 and R_2). Thus each of $P \cup R_1$ and $Q \cup R_2$ induced a cycle of $\overline{\overline{(K_t, w)}}_{2k+1}$ which uses less vertices of K_t than C. Furthermore, the total length of these two cycles is $|C| + 2k + 1$, hence one of the two is of length smaller than $2k+1$, this contradicts the choice of C, thus proving our claim. □

In the next theorem we show how to use the previous results to test whether a given weighted t-tree is $(2k+1)$-wide.

Theorem 3.6. *Let G be a t-tree with a clique-sequence $X_{t+1}, X_{t+2}, \ldots, X_{t+l}$. Let w be a weighting of G such that each X_i together with weights induced by w is $(2k+1)$-wide. Then (G, w) is $(2k+1)$-wide.*

Proof. We prove our claim by induction on l. If $l = 1$, then $(G, w) = (X_{t+1}, w)$ and thus (G, w) is $(2k+1)$-wide. Suppose (G', w) obtained by clique-sequence $X_{t+1}, X_{t+2}, \ldots, X_{t+l-1}$ is $(2k+1)$-wide, and let (G, w) be the graph obtained by adding a vertex v joined to t vertices $u_1, u_2, \ldots, u_t$. Thus $u_1, u_2, \ldots, u_t$ together with v induces X_{t+l}. Furthermore, suppose vu_i is of weight a_i $(i = 1, 2, \ldots, t)$. Since (G', w) is $(2k+1)$-wide, in order to prove that (G, w) is $(2k+1)$-wide, we only need to consider cycles of $\overline{\overline{(G, w)}}_{2k+1}$ which contains v and connects it to two vertices, say u_1, u_2, among $u_1, u_2, \cdots, u_t$ by paths of length a_i or $2k+1-a_i$, $i = 1, 2$. Let C be such a cycle, and assume by contradiction that it is an odd-cycle of length smaller than $2k+1$. Let p be the length of $u_1 v u_2$ portion of C and let p' be the length of the complementary part. Thus p and p' are of different parity. Suppose the $(2k+1)$-cycle connecting u_1 and u_2 in $\overline{\overline{(G, w)}}_{2k+1}$ is separated by u_1 and u_2 into two paths of length a and a', thus $a + a' = 2k+1$. By symmetry of a and a', assume $p + a$ is odd, then $p' + a'$ is also odd. Observe that each of $p + a$ and $p' + a'$ corresponds to the length of a cycle in $\overline{\overline{(G, w)}}_{2k+1}$. Since the sum of the lengths of these two cycles, $(|C| + 2k + 1)$, is less than $2(2k+1)$, one of them is of length smaller than $2k+1$. But $p + a$ being smaller than $2k+1$ will contradict the assumption of the theorem that each X_i together with induced weight by w is $(2k+1)$-wide, and $p' + a'$ being smaller than $2k+1$ contradicts our inductive assumption that (G', w) is $(2k+1)$-wide. These contradictions complete our proof. □

Combining these few results we have the following test for a t-tree to be $(2k+1)$-wide.

Corollary 3.7. *A weighted t-tree (T, w), $t \geq 2$, is $(2k+1)$-wide if and only if all induced weighted triangles of it are $(2k+1)$-wide.*

3.1. Partial and k-partial distance graphs

In this work, of the weighted graphs of previous discussion we will only consider the ones where the weight function is a specific metric function defined below.

Let G be a connected graph on n vertices. A *complete distance graph* of G is the weighted graph (K_n, d_G) where K_n is the complete graph on $V(G)$ and weight of an edge uv is the distance in G of u and v. Thus edges of weight 1 induce G. Any spanning (weighted) subgraph of (K_n, d_G) will then be referred to as a *partial distance graph* of G or a partial G-distance graph. Thus a partial G-distance graph (H, d_G) has $V(G)$ as its vertex set where each edge uv of H is given the weight $d_G(u, v)$, noting that H may not necessarily contain all edges of G. If, furthermore, we have $d_G(u, v) \leq k$ for each edge uv of H, then we say (H, d_G) is a *k-partial distance graph* of G or a k-partial G-distance graph.

A special family of k-partial G-distance graphs is built as follows:

Definition 3.8. Let G be a t-tree and let w be an edge-weighting with weights from $\{1, 2, \ldots, k\}$ such that (G, w) is $(2k+1)$-wide. Recall that we have associated with (G, w) a graph $\overline{\overline{(G, w)}}_{2k+1}$ whose vertices contain vertices of G and, furthermore, an edge of G is also an edge of $\overline{\overline{(G, w)}}_{2k+1}$ if and only if it is of weight 1. Let $G^*_{(w,2k+1)}$ be the weighted graph obtained from $\overline{\overline{(G, w)}}_{2k+1}$ by adding to it all edges of G,

then edges of $\overline{\overline{(G,w)}}_{2k+1}$ are assigned weight 1 and each other edge e is assigned the weight $w(e)$. As common edges are of weight 1 this works fine.

An example is given in Figure 2. The figure on the right is a weighted graph built from a weighted K_4, the weight of bold edges are the same as corresponding edges of (K_4, w) in the left and all other edges are of weight 1.

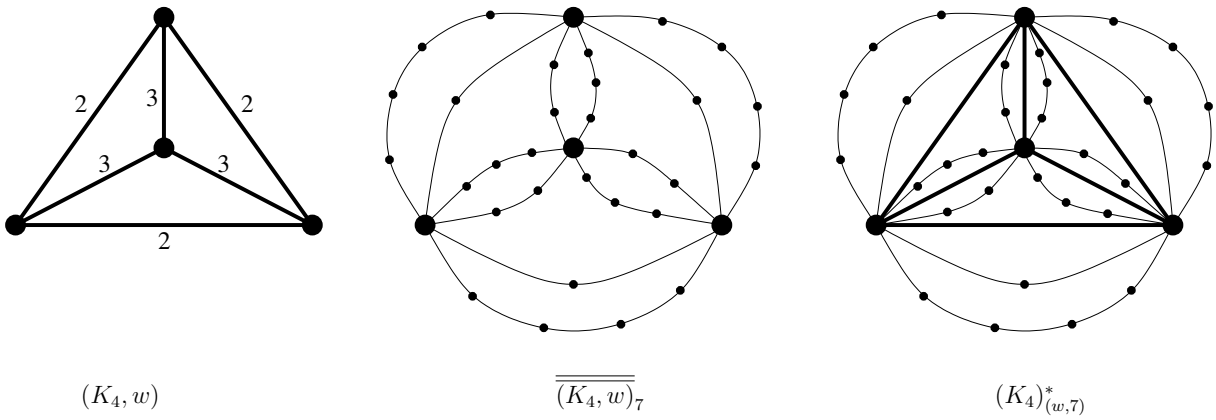

Figure 2: Examples of $\overline{\overline{(G,w)}}_{2k+1}$ and $G^*_{(w,2k+1)}$

As claimed in the following proposition, the condition that (G, w) is $(2k+1)$-wide implies that the extension of (G, w) into $G^*_{(w,2k+1)}$ transforms the general weight function w into a k-partial distance function (of $\overline{\overline{(G,w)}}_{2k+1}$). We leave the verification of this claim as an exercise.

Proposition 3.9. *Given a t-tree G and an edge-weighting w with weights from $\{1, 2, \ldots, k\}$ such that (G, w) is $(2k+1)$-wide, the weighted graph $G^*_{(w,2k+1)}$ is a k-partial $\overline{\overline{(G,w)}}_{2k+1}$-distance graph.*

3.2. Odd-girth and wideness

Here we present that, in essence of it, for a partial distance graph being $(2k+1)$-wide is stating that the reference graph is of odd-girth $2k+1$.

Lemma 3.10. *If B is of odd-girth $2k+1$, then any partial B-distance graph $\hat{B}$ is $(2k+1)$-wide.*

Proof. By the definition of a weighted graph to be $(2k+1)$-wide, we only need to prove that the complete B-weighted graph $\tilde{B}$ is $(2k+1)$-wide. To do this we need to show that any odd-cycle $\tilde{C}$ of $\overline{\overline{(\tilde{B},w)}}_{2k+1}$ has length at least $2k+1$. If $\tilde{C}$ has exactly two vertices from B then the length of $\tilde{C}$ is exactly $2k+1$ by the construction. Assume $\tilde{C}$ at least three vertices from B, denote them $x_1, x_2, \ldots, x_t$ in the clockwise orientation of $\tilde{C}$. Denote the distance of $x_i x_{i+1}$ in $\tilde{C}$ by l_i. If there exist l_i and l_j such that $l_i, l_j \geq k$ then $\sum_{s=1}^{t} l_s \geq 2k+1$. So the length of $\tilde{C}$ is at least $2k+1$. Suppose that $l_i < k$ for any $i = 1, 2, \ldots, t$. Then we know that $l_i = d_B(x_i, x_{i+1})$, so there is a closed-walk of odd length going through x_1 to x_t in B whose length is $\sum_{s=1}^{t} l_s$. Since B is of odd-girth $2k+1$, therefore, $\sum_{s=1}^{t} l_s \geq 2k+1$. So the length of $\tilde{C}$ is at least $2k+1$. Finally consider the case that there is exactly one l_i with $l_i \geq k$, by the symmetry we may assume $l_1 \geq k$. Hence, $l_i = d_B(x_i, x_{i+1})$ for $i = 2, \ldots, t$. We classify into two subcases. If $l_1 = k$, then we have $l_1 = w(x_1, x_2) = k \leq d_B(x_1, x_2)$. By the triangular inequality, we get $l_2 + \cdots + l_t \geq d_B(x_1, x_2)$. So $\sum_{s=1}^{t} l_s \geq k + d_B(x_1, x_2) \geq 2k$. Since $\sum_{s=1}^{t} l_s$ is odd so $\sum_{s=1}^{t} l_s \geq 2k+1$. If $l_1 \geq k+1$, then $l_1 = 2k+1-w(x_1, x_2)$.

Hence, $\sum_{s=1}^{t} l_s = 2k+1-w(x_1,x_2)+l_2+\cdots+l_t$. Since $w(x_1,x_2) \le d_B(x_1,x_2)$, by triangular inequality, we have $\sum_{s=1}^{t} l_s \ge 2k+1$. We conclude that $\tilde{C}$ has length at least $2k+1$ in each case. □

The following theorem then is a key tool of our work. It claims, basically, that a homomorphism of a graph G to a graph H where both G and H are of odd-girth $2k+1$ can be viewed as a mapping which preserves more than just adjacency. It is only a restatement of Lemma 1.2 using the terminology we have just developed.

Theorem 3.11. *Let G and H be two graphs of odd-girth $2k+1$. Let (G', w_1) and (H', w_2) be, respectively, partial G-distance and partial H-distance graphs such that the end points of any edge of weight at least 2 of (G', w_1) (resp. (H', w_2)) is in a $(2k+1)$-cycle of G (resp. H). Then any homomorphism of G to H is also a homomorphism of (G', w_1) to (H', w_2).*

3.3. List of all $(2k+1)$-wide complete graphs on $t+1$ vertices

Given positive integers k and t and toward working with graphs in $\mathcal{PT}_{t,2k+1}$ we will need list of all weighting w of the complete graphs on $t+1$ vertices such that (K_{t+1}, w) is $(2k+1)$-wide. As k and t are fixed numbers, this list can be easily computed. The computation of this list is further simplified by Theorem 3.5. We denote the list of such weights by $L(t+1, 2k+1)$. The total number of weightings on a K_{t+1} whose vertices are labeled, with weight of each edge being an integer between 1 and k, is $k^{\binom{t+1}{2}}$. This provides an upper bound on the number of elements of $L(t+1, 2k+1)$. This upper bound is not too far from the exact number of elements by the following observation:

Proposition 3.12. *If w is a weighting of K_{t+1} satisfying $\frac{2k}{3} \le w(e) \le k$, then (K_{t+1}, w) is $(2k+1)$-wide.*

Proof. By Theorem 3.5 it is enough to check that for each triangle the induced weighted triangle is $(2k+1)$-wide. Let a, b, c be weights of a triangle in (K_{t+1}, w). We use Proposition 3.3 to test if this triangle is $(2k+1)$-wide. If $a+b+c$ is odd, then it is at least $2k+1$. If it is even, then $a+b \ge \frac{4k}{3}$ but $c \le k$, which implies $a+b \ge c$. □

Thus given k and t, there are at least $\frac{1}{\binom{t+1}{2}!}(\frac{k}{3})^{\binom{t+1}{2}}$ non isomorphic elements in $L(t+1, 2k+1)$. We leave finding the exact number of elements of $L(t+1, 2k+1)$ as an open question. To help with a better understanding of the notion in Figure 3 we have provided the full list of 7 non-isomorphic elements of $L(4, 5)$.

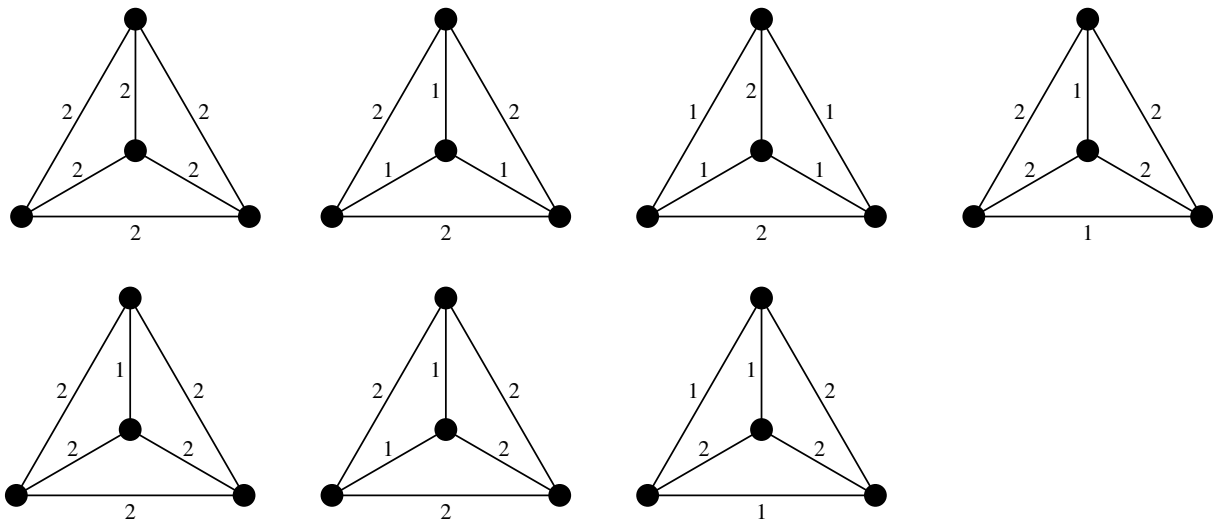

Figure 3: The list $L(4, 5)$ of all 5-wide K_4's

In the next lemma we show that our list $L(t+1, 2k+1)$ maintains a weak notion of monotonicity. That is to say if in a given $(2k+1)$-wide weighting all entries corresponding to a vertex u are raised to the maximum (which is k), then the result will still be a member of our list.

Lemma 3.13. *Let w be a weighting of K_{t+1} such that (K_{t+1}, w) is $(2k+1)$-wide. If we change the weight of each of the t edges incident to a same vertex to k, then the resulting weighted graph (K_{t+1}, w') is still $(2k+1)$-wide.*

Proof. By the assumption, we have $w'(e) \leq k$ for any $e \in E(K_{t+1})$. In order to prove that (K_{t+1}, w') is $(2k+1)$-wide, by the definition of $\overline{\overline{(K_{t+1}, w')}}_{2k+1}$, we only need to prove that every odd-cycle C which contains at least three vertices of K_{t+1} has length at least $2k+1$. Let v_0 be the vertex which we change the weights of its adjacent edges, if $v_0 \in V(C)$ then the length of C is at least $2k+1$. Otherwise, C is an odd-cycle of $\overline{\overline{(K_{t+1}, w)}}_{2k+1}$. Hence, the length of C is at least $2k+1$. □

4. $(t, 2k+1)$-closed sets of cliques

We are now ready to give our main definition, using which, we can determine whether a given graph B of odd-girth $2k+1$ bounds $\mathcal{PT}_{t,2k+1}$.

Definition 4.1. A collection $\mathcal{W}$ of $(t+1)$-cliques of a $(2k+1)$-wide weighted graph (G, w) is $(t, 2k+1)$-*closed* if it satisfies the following: for any $(t+1)$-clique K_{t+1} of $\mathcal{W}$, for any t-subclique of K_{t+1} on vertices $v_1, v_2, \ldots, v_t$, suppose $a_1, a_2, \ldots, a_t$ are positive integers satisfying $a_i \leq k$ $(i = 1, 2, \ldots, t)$ and such a $(t+1)$-clique obtained from the induced (weighted) t-clique on $v_1, v_2, \ldots, v_t$ and a vertex v such that vv_i is of weight a_i is $(2k+1)$-wide. Then there exists a vertex $y \in V(G)$ such that $y, v_1, v_2, \ldots, v_t$ induces a member of $\mathcal{W}$ and the weight of yv_i is a_i. Given such a set $\mathcal{W}$ of $(t+1)$-cliques, the set of t-cliques induced by cliques in $\mathcal{W}$ is denoted by $\mathcal{W}'$.

For a better understanding of this definition we use the list of Figure 3. Suppose (G, w) is a weighted graph which is 5-wide (in a sense it is triangle-free), and let $\mathcal{W}$ be a collection of its 4-cliques which is $(3, 5)$-closed. Let W_1 be a member of $\mathcal{W}$ on vertices x, y, z, t and suppose all edges are of weight 2, i.e., it is isomorphic to the first weighted graph in the list of Figure 3. Consider the triangle induced by xyz. That is a triangle all whose edges are of weight 2. There are three other weighted K_4 which contain an isomorphic copy of the triangle xyz, the second, fifth and sixth graphs in the list. Considering the first, we conclude that the existence of a vertex u_{111} which is adjacent to x, y and z (with weight 1) such that the complete graph induced by x, y, z and u_{111} is also in the list. The fifth element implies existence of 3 such vertices and cliques: u_{122} is vertex which is adjacent to x (with weight 1) and to y and z both with edges of weight 2, similarly u_{212} and u_{221} are defined, noting that each of the three cliques induced by $\{x, y, z, u_{122}\}$, $\{x, y, z, u_{212}\}$, $\{x, y, z, u_{221}\}$ must be in $\mathcal{W}$. Similarly, there must be three other vertices u_{112}, u_{121} and u_{211} giving cliques isomorphic to the sixth element of the list. We note that each of these vertices might be used again to satisfy the condition for other triangles of W_1 or for elements of $\mathcal{W}$. All together, as we will see later, the smallest 5-wide weighted graph with a nontrivial $(3, 5)$-closed set of 4-cliques has 16 vertices.

A $(t, 2k+1)$-closed set of cliques of a weighted graph must then have a large number of $(t+1)$-cliques. To find smallest such a set would probably address challenging questions. An easy lower bound on the number of elements of such a set is the number of elements of $L(t+1, 2k+1)$, more precisely given a $(t, 2k+1)$-closed set of cliques if it is nonempty, then it must have one $(t+1)$-clique for each element of $L(t+1, 2k+1)$ as we prove in the following theorem.

Theorem 4.2. *Given a $(2k+1)$-wide weighted graph (G, w), if a nonempty collection $\mathcal{W}$ of its $(t+1)$-cliques is $(t, 2k+1)$-closed, then for any weighting w of K_{t+1} such that $w \in L(t+1, 2k+1)$ there is an isomorphic copy of (K_{t+1}, w) in $\mathcal{W}$.*

Proof. Since $\mathcal{W}$ is nonempty, there exists an element (K_{t+1}, w_0) in $\mathcal{W}$. Let $V(K_{t+1}) = \{v_0, v_1, \ldots, v_t\}$, then $(K_{t+1}/\{v_0\}, w_0) \in \mathcal{W}'$. If we change the weights of the edges adjacent to v_0 to k, by Lemma 3.13 the resulting $(t+1)$-clique is $(2k+1)$-wide. As $\mathcal{W}$ is $(t, 2k+1)$-closed, there exists a vertex u_0 which is adjacent to each of $v_1, v_2, \ldots, v_t$ with an edge of weight k and that the $(t+1)$-clique induced by $u_0, v_1, v_2, \ldots, v_t$ is in $\mathcal{W}$. By repeating this process we may replace each v_i with a vertex u_i which is adjacent to the vertices

of the current clique we are working with edges of weight k. Thus at final step we have a clique in $\mathcal{W}$ all whose edges are of weight k.

In summary, starting with any $(2k+1)$-wide $(t+1)$-clique in $\mathcal{W}$ we get that there is a clique in $\mathcal{W}$ all whose edges are of weight k. However, this is a reversible construction, and starting with one such a clique (where all edges are of weight k) and using the property of being $(t, 2k+1)$-closed, we can get an isomorphic copy of any other member of $L(t+1, 2k+1)$ in $\mathcal{W}$. □

4.1. Necessary and sufficient condition

Recall that $\mathcal{PT}_t$ is the class of partial t-trees, and $\mathcal{PT}_{t,2k+1}$ is the subclass of partial t-trees of odd-girth at least $2k+1$. We are now ready to state and prove our main theorem.

Theorem 4.3. *A graph B of odd-girth $2k+1$ admits a homomorphism from every partial t-tree of odd-girth at least $2k+1$ if and only if there is a partial B-distance graph $(\hat{B}, w)$ with a nonempty set $\mathcal{W}$ of $(t+1)$-cliques of $(\hat{B}, w)$ which is $(t, 2k+1)$-closed.*

Proof. First, we prove the sufficient part. Assume such a weighted graph $\hat{B}$ and a nonempty set $\mathcal{W}$ of $(t+1)$-cliques exist. Suppose G is a partial t-tree of odd-girth at least $2k+1$. Let $\tilde{G}$ be a t-tree containing G as a subgraph. We form a weighted graph on $\tilde{G}$ by defining the weight function $\varphi : E(\tilde{G}) \to \{1, 2, \ldots, k\}$ by $\varphi(xy) = \min\{d_G(x, y), k\}$ for any $xy \in E(\tilde{G})$. By Lemma 3.10, we know that $(\tilde{G}, \varphi)$ is a $(2k+1)$-wide weighted graph. Let $X_1, X_2, \ldots, X_s$ be a $(t+1)$-clique-sequence from which $\tilde{G}$ is built. Since X_1 is a subgraph of $\tilde{G}$, we know that (X_1, φ) is $(2k+1)$-wide. By Theorem 4.2, there is an isomorphic copy of X_1 in $\mathcal{W}$, let C_1 be this copy. Let h be an isomorphism of X_1 to C_1. Observe that in particular h is a homomorphism of X_1 to C_1. Our goal is to extend h as a homomorphism of $\tilde{G}$ to $\hat{B}$. Let $V(X_2) = \{x_1, x_2, \ldots, x_t, x_{t+1}\}$, by the definition of partial t-tree, and without loss of generality, assume $x_1, x_2, \ldots, x_t \in V(X_1)$. Since $\mathcal{W}$ is $(t, 2k+1)$-closed and X_2 is $(2k+1)$-wide, there exists $y \in V(B)$ such that $\{y, h(x_1), h(x_2), \ldots, h(x_t)\}$ induces a member of $\mathcal{W}$ and the weight of $yh(x_i)$ is the same as the weight of $\varphi(x_{t+1}x_i)$ for any $i \in \{1, 2, \ldots, t\}$. Let $h(x_{t+1}) = y$, hence, this extended h is a homomorphism of $(X_1 \cup X_2, \varphi)$ to $\hat{B}$. By continuing this process, finally we extend h to be a homomorphism of $(\tilde{G}, \varphi)$ to $(\hat{B}, w)$. Thus for any edge xy of $\tilde{G}$ which is of weight 1, $h(x)h(y)$ is an edge of $\hat{B}$ of weight 1. That is to say $h(x)h(y)$ is an edge of B. Therefore, h is also a homomorphism of G to B.

Next, we want to prove that the condition is necessary.

We first define $\hat{B}$ as follows: vertices of $\hat{B}$ are the same as vertices of B. A pair x, y of vertices is an edge of $\hat{B}$ if they belong to a common $(2k+1)$-cycle of B. Thus $\hat{B}$ does not necessarily contains all edges of B. We then define the weight φ of an edge xy of $\hat{B}$ to be the distance of the pair in B. That is the same as their distance in the odd-cycle containing them since B is of odd-girth $2k+1$. Our aim is to show that $(\hat{B}, \varphi)$ admits a non-empty set $\mathcal{W}$ of $(t+1)$-cliques which is $(t, 2k+1)$-closed.

Recall that given a t-tree G and a weighting w $(1 \leq w(e) \leq k)$ of its edges we have associated with (G, w) a weighted graph $G^*_{(w,2k+1)}$ which, by Proposition 3.9, is a partial distance graph of $\overline{\overline{(G, w)}}_{2k+1}$. Observe that for $t \geq 2$ the underlying graph of $G^*_{(w,2k+1)}$ is a partial t-tree. Since $\overline{\overline{(G, w)}}_{2k+1}$ is a partial t-tree of odd-girth $2k+1$, by our assumption, it admits a homomorphism to B. Thus by Theorem 3.11, $G^*_{(w,2k+1)}$ maps to $(\hat{B}, \varphi)$.

Let $\mathcal{PT}^*_{t,2k+1}$ be the set of weighted graphs $G^*_{(w,2k+1)}$ where G is a t-tree such that, together with edge-weighting w, (G, w) is $(2k+1)$-wide. As mentioned above each member of $\mathcal{PT}^*_{t,2k+1}$ admits a homomorphism to $(\hat{B}, \varphi)$. Observe that in a mapping of any such member $G^*_{(w,2k+1)}$ to $(\hat{B}, \varphi)$ the image of any $(t+1)$-clique of $G^*_{(w,2k+1)}$ is a weighted $(t+1)$-clique in $(\hat{B}, \varphi)$. Let $\mathcal{W}$ be a minimal set of $(t+1)$-cliques in $(\hat{B}, \varphi)$ with the following property: any member $G^*_{(w,2k+1)}$ of $\mathcal{PT}^*_{t,2k+1}$ admits a homomorphism to $(\hat{B}, \varphi)$ such that image of any $(t+1)$-clique of $G^*_{(w,2k+1)}$ is a member of $\mathcal{W}$. We will denote the existence of such a homomorphism by $G^*_{(w,2k+1)} \to (\hat{B}, \varphi, \mathcal{W})$.

We claim that $\mathcal{W}$ satisfies the condition of our theorem, i.e., $\mathcal{W}$ is not empty and it is $(t, 2k+1)$-closed.

To see that $\mathcal{W}$ is not empty it is enough to take any $(2k+1)$-wide K_{t+1} as (G, w) and consider the corresponding $G^*_{(w,2k+1)}$.

The minimality of $\mathcal{W}$ means that given a $(t+1)$-clique $C_1 \in \mathcal{W}$, there exists $\hat{G}_{C_1} \in \mathcal{PT}^*_{t,2k+1}$ such that for any homomorphism of $(\hat{G}_{C_1}, w)$ to $(\hat{B}, \varphi, \mathcal{W})$, some $(t+1)$-clique of $(\hat{G}_{C_1}, w)$ will be mapped to C_1.

To prove that $\mathcal{W}$ is $(t, 2k+1)$-closed, we take a $(t+1)$-clique, say C_1, we take t vertices $v_1, v_2, \ldots, v_t$ from C_1 and assume adding a vertex x with $\varphi(xv_i) = a_i$ produces a $(t+1)$-clique which is $(2k+1)$-wide. Then we need to show that there exists a vertex $v \in V(B)$ such that $v_1, \ldots, v_t, v$ induces a $(t+1)$-clique in $\mathcal{W}$ and such that $\varphi(vv_i) = a_i$ for $i = 1, \ldots, t$. Consider all isomorphic copies of C_1 in $\hat{G}_{C_1}$. For each such copy K and for each isomorphism, add a vertex x_K then join it to all copies $v_i's$ with an edge weighted a_i to the vertex corresponding to v_i. Also add two paths of length a_i and $2k+1-a_i$ between x_K and the vertex corresponding to v_i. Each edge of these two paths is of weight 1. Let G^* be the new weighted graph. By the construction and by Proposition 3.9 the weighted graph G^* is a member of $\mathcal{PT}^*_{t,2k+1}$, and, therefore, there exists a homomorphism ρ of G^* to $(\hat{B}, \varphi, \mathcal{W})$. Observe that $\hat{G}_{C_1} \subseteq G^*$, thus ρ induces a homomorphism of $\hat{G}_{C_1}$ to $(\hat{B}, \varphi, \mathcal{W})$. By the choice of $\hat{G}_{C_1}$, some $(t+1)$-clique of $\hat{G}_{C_1}$, say K is mapped to C_1. The image of x_K together with $v_1, v_2, \ldots, v_t$ is a $(t+1)$-clique in $\mathcal{W}$ that we are looking for. In other words, $\rho(x_K)$ can be v. Therefore, $\mathcal{W}$ is $(t, 2k+1)$-closed. □

5. Projective cubes and bounding partial 3-trees

Theorem 4.3 is an extension of a necessary and sufficient condition provided in [2] to test if a given graph of odd-girth $2k+1$ bounds the class of K_4-minor-free graphs of odd-girth at least $2k+1$. The question of finding smallest such a graph is discussed in that paper where it is shown that order of a smallest such graph must be $\theta(k^2)$. One naturally expects that the analogous question would become more difficult for $\mathcal{PT}_{t,2k+1}$ for $t \geq 3$. While generally this might be the case, in this section we show that the case $t = 3$ is rather special by providing the optimal answer. Here we give a graph of odd-girth $2k+1$ on 2^{2k} vertices which bounds the class $\mathcal{PT}_{3,2k+1}$ and we point out that there can be no such graph on less number of vertices. Our results here can be viewed as strongest support so far provided for the general case of a conjecture in extension of the four-color theorem. This conjecture will be restated after we introduce the necessary definition.

We first recall the definition of a Cayley graph: let Γ be an additive group and let S be a symmetric subset of Γ (i.e., for each $x \in S$ we have $-x \in S$) and furthermore suppose that 0 is not a member of S. Then the Cayley graph (Γ, S) is the graph whose vertices are elements of Γ where two vertices x and y are adjacent if $x - y \in S$. That S is symmetric implies that G is a graph and not a digraph. As we will consider only binary groups here, we have $-x = x$ for each x and thus all sets are symmetric here. The important advantage of considering Cayley graphs on binary groups is the naturally associated edge-coloring. Given a Cayley graph on a binary group to each edge uv we assign $u+v = u-v = v-u \in S$.

The main targets of this section are the following Cayley graphs on the binary group $\mathbb{Z}_2^k$.

Definition 5.1. The *projective cube* of dimension k is the Cayley graph $(\mathbb{Z}_2^k, \{e_1, e_2, \ldots, e_k, J\})$ where the e_i's are the vectors of the standard basis and J is the all-1 vector.

Recall that the Cayley graph $(\mathbb{Z}_2^k, \{e_1, e_2, \ldots, e_k\})$ is the well known hypercube $H(k)$, where graph distances are the same as hamming distances. Therefore $PC(k)$ is built from $H(k)$ by adding an edge between antipodal pairs. It is not hard to see that this graph is also built from $H(k+1)$ by identifying antipodal pairs, it is this projection that has given our choice of name "projective cube" to these graphs, however, they are referred to as "folded cube" more often. It can be checked that $PC(2k-1)$ is bipartite. On the other hand, $PC(2k)$ is a 4-chromatic graph of odd-girth $2k+1$. It is easily verified that $PC(2)$ is isomorphic to K_4, thus the following conjecture of the second author from [14] is an extension of the four-color theorem:

Conjecture 5.2. *Every planar graph of odd-girth at least $2k+1$ admits a homomorphism to $PC(2k)$.*

The conjecture is believed to be true for the larger class of K_5-minor-free graphs, (see [16]). Even further extension using notion of signed graphs is considered, we refer to [16] for more details. The conjecture is about finding an optimal bound in the following sense: it is proved in [18] that if true, then $PC(2k)$ is a smallest graph (both in terms of the number of edges and the number of vertices) of odd-girth $2k+1$ which bounds the class of all planar graphs of odd-girth at least $2k+1$. That is to say, if B is a graph of odd-girth $2k+1$ to which every planar graph of odd-girth at least $2k+1$ admits a homomorphism, then B has at least as many vertices and as many edges as $PC(2k)$.

The case $k = 1$ of the conjecture is just the four-color theorem, and its extension to the K_5-minor-free graphs is obtained by Wagner's theorem on decomposition of K_5-minor-free graphs. The case $k = 2$ is

proved in [14] (using the four-color theorem) and extension to K_5-minor-free graphs is given in [19]. The case $k = 3$ follows from recent results on edge-coloring of planar multigraphs also based on the four-color theorem (see [16]).

For lager values of k the conjecture remains open, and the best known support was a verification for the class of K_4-minor-free graphs proved in [2]. Here, as an application of our work, we show that the conjecture holds for the larger class of partial 3-trees of odd-girth at least $2k + 1$. We note that this is a subclass of K_5-minor-free graphs, but it does contain some non-planar graphs. Thus our result here provides the strongest evidence so far provided in support of the conjecture. To proceed we need to establish some notations and prove some properties of $PC(2k)$. To this end we will have to provide proofs for some already known facts about projective cubes and then strengthen such results.

We start by recalling a general fact about Cayley graphs on binary groups. Let G be Cayley graph on a binary group and let ϕ be the associated edge-coloring. Suppose $v_1, v_2, \ldots, v_l, v_1$ is a closed-walk of G (with this order of vertices). Then $\phi(v_1v_2) + \phi(v_2v_3) + \cdots + \phi(v_lv_1) = 2(v_1 + v_2 + \cdots + v_l) = 0$. In particular this is the case for all cycles.

Consider the set $S_{2k} = \{e_1, e_2, \ldots, e_{2k}, J\}$. The only linear relations among elements of S_{2k} are $2x = 0$ for any x and $\sum_{x \in S_{2k}} x = 0$. This fact together with the previous observation partitions the set of cycles of $PC(2k)$ into two types: *i.* cycles C where $\phi(v_1v_2) + \phi(v_2v_3) + \cdots + \phi(v_lv_1) = 0$ because every element of S_{2k} appears an even number of times as color on edges of C (0 as an even number also allowed), and *ii.* cycles C where $\phi(v_1v_2) + \phi(v_2v_3) + \cdots + \phi(v_lv_1) = \sum_{x \in S_{2k}} x = 0$, that is to say each and every element of S_{2k} appears as color on edges of C an odd number of times. As S_{2k} has an odd number of elements, this second type of cycles then are the class of odd-cycles of $PC(2k)$ and the first type are exactly the class of even-cycles. As a consequence we find that $PC(2k)$ has odd-girth $2k + 1$. But in fact a stronger statement follows based on the following notation:

Let u and v be two vertices of $PC(2k)$ and let P be a shortest $u - v$ path, let S_{uv} be the set of the colors of edges of P, thus $S_{uv} \subset S_{2k}$.

Theorem 5.3. *Given $PC(2k)$ and any two vertices u and v, the set S_{uv} corresponding to the colors of the edges of a shortest $u-v$ path P is independent of the choice of P and thus well defined. Furthermore, given a $u-v$ path P' of length $2k+1-d(u,v)$ the set of colors of the edges of P' is $\bar{S}_{uv} = S_{2k} \setminus S_{uv}$ and thus independent of the choice of P'.*

Proof. In fact the definition in each part provides the proof for the claim of the other part: choose a $u - v$ path P of length $d(u, v)$ and let S_{uv} be the set of colors of the edges of P. For an arbitrary choice P' of a $u - v$ path of length $2k + 1 - d(u, v)$ consider the closed-walk which traverses P and then P'. There are an odd number of edges on this walk, $2k + 1$ edges to be exact, and sum of their colors is 0. The only way for this to happen is that the edges of P' are colored by the elements of $S_{2k} \setminus S_{uv}$. For the second part we should note that at least one path of length $2k + 1 - d(u, v)$ exists. Simply consider the set $\bar{S}_{uv}$ (which we now know is well defined), order its elements as $x_1, x_2, \ldots, x_{2k+1-d(u,v)}$ and consider the following sequence of vertices: $u, u + x_1, u + x_1 + x_2, \ldots, u + x_1 + x_2 + \cdots + x_{2k+1-d(u,v)} = v$. Now by choosing one such path, and as in the previous part we may show that S_{uv} is well defined. □

This leads to the following labeling of the edges of the complete graph on 2^{2k} vertices using $PC(2k)$.

Definition 5.4. The *$PC(2k)$-edge-labeled complete graph* is the complete graph on vertices of $PC(2k)$ where edge uv is labeled by the partition $\{S_{uv}, \bar{S}_{uv}\}$.

Observe that replacing each such label with the order of S_{uv} (that is the smaller of the two parts) we have the complete $PC(2k)$-distance graph which is $(2k + 1)$-wide as $PC(2k)$ is of odd-girth $2k + 1$. Before we use this labeling to apply our main theorem on the complete $PC(2k)$-distance graph, we use it to prove a high level of symmetries in $PC(2k)$. To this end we will need the following property of the complete $PC(2k)$-distance graph.

Proposition 5.5. *Let x, y and z be three vertices of the $PC(2k)$-edge-labeled complete graph. Let $A \in \{S_{xy}, \bar{S}_{xy}\}$, $B \in \{S_{xz}, \bar{S}_{xz}\}$ and $C \in \{S_{yz}, \bar{S}_{yz}\}$. Depending on the parity of $|A| + |B| + |C|$ one of the following holds:*

- *Each element of S_{2k} appears in an even number of the sets A, B or C, that is to say either it appears in non of them, or exactly two of them (this is the case when $|A|+|B|+|C|$ is even).*
- *Each element of S_{2k} appears in an odd number of the sets A, B or C, that is to say it appears either in exactly one of them or in all three of them (this is the case when $|A|+|B|+|C|$ is odd).*

Proof. Observe that the choice of A corresponds to colors of the edges of an $x-y$ path P_A, similarly choice of B and C correspond to paths P_B and P_C between $x-z$ and $y-z$, respectively. Thus $|A|+|B|+|C|$ corresponds to the length of the closed-walk starting at x and traversing P_A, then P_C and then returning to x through P_B. Thus sum of the elements of A and B and C is 0. That means if a color comes an odd-number of times in the closed-walk, then so do all other colors, and thus $|A|+|B|+|C|$ is an odd number. Otherwise all colors appear an even number of times, which means $|A|+|B|+|C|$ is also even. □

Corollary 5.6. *Given subsets A, B and C of the previous proposition such that $|A|+|B|+|C|$ is odd, the number of elements that appears in all three of them is $\frac{1}{2}(|A|+|B|+|C|-(2k+1))$.*

This corollary is of importance for two main reasons: The first reason is that $|A|$, $|B|$ and $|C|$ are the edge weights of the weighted triangle induced by x, y and z of the complete $PC(2k)$-distance graph. Thus this triangle is $(2k+1)$-wide and the value $\frac{1}{2}(|A|+|B|+|C|-(2k+1))$ of this corollary corresponds to $f_{2k+1}(|A|,|B|,|C|)$ of Proposition 3.4. The second reason is that the number of elements appearing in all three of A, B and C is a function of the order of A, B and C and not the choice of A, B, C. This implies a very high level of symmetries as we will discuss below.

A graph G is said to be *distance transitive* if for two pairs (u,v) and (x,y) of vertices, if $d_G(u,v)=d_G(x,y)$, then there is an automorphism of G which maps u to x and v to y. Furthermore, G is said to be *triple transitive* if for two triples (u,v,w) and (x,y,z) of vertices, if $d_G(u,v)=d_G(x,y)$, $d_G(u,w)=d_G(x,z)$, $d_G(v,w)=d_G(y,z)$, then there is an automorphism ζ of G such that $\zeta(u)=x$, $\zeta(v)=y$ and $\zeta(w)=z$. In the next theorem we give a proof of a result of [13] (see also [3]) using the terminology we have developed here. That is to show that $PC(2k)$ is triple transitive.

Theorem 5.7. *The graph $PC(2k)$ is triple-transitive.*

Proof. In order to prove this we should introduce automorphisms of this graph. To this end, similar to the labeling of the edges of the complete $PC(2k)$-labeled graph on 2^{2k} vertices, and using this labeling of edges, we label the vertices of $PC(2k)$ also by partitioning of S_{2k}. To start with we label the vertex 0 by the trivial partition $\{\emptyset, S_{2k}\}$. Then the vertex u receives the label of the edge $0u$, that is $\{S_{0u}, \bar{S}_{0u}\}$. Using this labeling one observes easily that any permutation of S_{2k} is an automorphism of the complete $PC(2k)$-labeled graph and, therefore, also an automorphism of $PC(2k)$. In fact this is the full list of automorphisms that has 0 as its fixed point, but we do not need to prove this fact here. Given a set $A\subset\{e_1,e_2,\ldots,e_{2k}\}$ one may get an automorphism of complete $PC(2k)$-labeled graph (and thus of $PC(2k)$) by mapping the vertex labeled $\{X,\bar{X}\}$ to the vertex labeled $\{X\oplus A,\bar{X}\oplus A\}$ (if a is the vector in $\mathbb{Z}_2^{2k}$ whose support is A, then this automorphism corresponds to the mapping of $u\to u+a$). Unless A is the empty set, this automorphism never fixes the vertex 0. By composing two of the automorphisms described above one, of course, gets an automorphism of complete $PC(2k)$-labeled graph and thus of $PC(2k)$. This is indeed the full list of automorphisms of this graph, but we do not need to prove this stronger statement here either.

To prove the claim of the theorem, let $\{x,y,z\}$ and $\{u,v,w\}$ be two triples of vertices of $PC(2k)$ such that $d_{PC(2k)}(x,y)=d_{PC(2k)}(u,v)$, $d_{PC(2k)}(x,z)=d_{PC(2k)}(u,w)$, $d_{PC(2k)}(y,z)=d_{PC(2k)}(v,w)$. Note that by taking a symmetric difference one may easily map the vertex u to the vertex x. If v and w are mapped to v' and w' by this automorphism, then it is enough to prove that the triple x,v',w' can be mapped to x,y,z by an automorphism of $PC(2k)$, so we may assume $u=x$ in the rest of this proof.

The assumption of $d_{PC(2k)}(x,y)=d_{PC(2k)}(x,v)$, $d_{PC(2k)}(x,z)=d_{PC(2k)}(x,w)$ and $d_{PC(2k)}(y,z)=d_{PC(2k)}(v,w)$ implies that $|S_{xy}|=|S_{xv}|$, $|S_{xz}|=|S_{xw}|$ and $|S_{yz}|=|S_{vw}|$. Consider the triangle induced by the vertices x,y,z in the complete $PC(2k)$-labeled graph, the edges of this triangle are labeled by $\{S_{xy},\bar{S}_{xy}\}$, $\{S_{xz},\bar{S}_{xz}\}$ and $\{S_{yz},\bar{S}_{yz}\}$. Let $A\in\{S_{xy},\bar{S}_{xy}\}$, $B\in\{S_{xz},\bar{S}_{xz}\}$, and $C\in\{S_{yz},\bar{S}_{yz}\}$ be such that $|A|+|B|+|C|$ is odd. Noting that $|S_{xy}|\le k<\bar{S}_{xy}$, let A' be the element of $\{S_{xv},\bar{S}_{xv}\}$ which is of the same order as A, similarly define B' and C'. To complete the proof it would be enough to find a permutation of S_{2k} which maps A to A', B to B' and C to C'.

Thanks to Corollary 5.6, we know A, B, C have $\frac{1}{2}(|A|+|B|+|C|-(2k+1))$ common elements (each appearing in all three of them) and every other element of S_{2k} appears exactly in one of the three. The same holds for A', B' and C'. As $|A|=|A'|$, $|B|=|B'|$ and $|C|=|C'|$ the triple set A, B, C has as many common elements as A', B', C'. Thus it is enough (and necessary) to choose a permutation of S_{2k} which maps common elements of A, B, C to common elements of A', B', C'. □

We are now ready to prove the main theorem of this section:

Theorem 5.8. *The set $\mathcal{W}$ of 4-cliques of the complete $PC(2k)$-distance graph is $(3, 2k+1)$-closed.*

Proof. Since the projective cube $PC(2k)$ is triple-transitive, we only need to prove that for each $(2k+1)$-wide weighted graph (K_4, w) there is an isomorphic copy in $\mathcal{W}$.

Let (K_4, w) be a $(2k+1)$-wide graph with vertices x, y, z, u. Suppose that $w(xy)=a', w(xz)=b', w(xu)=c', w(zu)=a, w(yu)=b, w(yz)=c$. If necessary, we change the weight of an edge to its complementary weight $2k+1-w$ to make sure that the elements of each pair $\{a,a'\},\{b,b'\},\{c,c'\}$ are of the same parity and the sum $a+b+c$ is odd. Therefore, the weight sum of the four triangles of (K_4, w) are $a+b+c$, $a+b'+c'$, $a'+b+c'$ and $a'+b'+c$, each being an odd number. Let $f_x=\frac{1}{2}(a+b+c-2k-1), f_y=\frac{1}{2}(a+b'+c'-2k-1), f_z=\frac{1}{2}(a'+b+c'-2k-1), f_u=\frac{1}{2}(a'+b'+c-2k-1)$. By Theorem 3.5 and Proposition 5.5, we know that each of the four values $f_v, v\in\{x,y,z,u\}$, is a nonnegative integer smaller than the minimum of the three elements defining it.

Next, we will assign a subset $A_{vv'}$ of $S_{2k}=\{e_1,e_2,\ldots,e_{2k},J\}$ to each edge vv' of K_4 such that $|A_{vv'}|$ is either $w(vv')$ or $2k+1-w(vv')$ and for each triangle tvw of K_4, each element of $\{e_1,e_2,\ldots,e_{2k},J\}$ appears either in exactly one of A_{tv}, A_{tw}, A_{vw} or in all of them in order to satisfy Proposition 5.5 (here, t, v, w can be any triple among x,y,z,u, similarly v, v' can be any pair of these four vertices).

Without loss of generality, assume that $f_x=\min\{f_x,f_y,f_z,f_u\}$. Let S_x be a subset of $S_{2k}=\{e_1,e_2,\ldots,e_{2k},J\}$ of order f_x. We partition the set $S_{2k}\setminus S_x$ into three disjoint subsets S_{yz}, S_{yu}, S_{zu} of order $|S_{yz}|=c-f_x, |S_{yu}|=b-f_x, |S_{zu}|=a-f_x$. Define $A_{yz}=S_{yz}\cup S_x, A_{yu}=S_{yu}\cup S_x, A_{zu}=S_{zu}\cup S_x$. Let S'_y, S'_z, S'_u be subsets such that $S'_y\subset S_{zu}$ and $|S'_y|=f_y-f_x$, $S'_z\subset S_{yu}$ and $|S'_z|=f_z-f_x$, $S'_u\subset S_{yz}$ and $|S'_u|=f_u-f_x$. The set $S_y=S'_y\cup S_x$ is of order f_y, $S_z=S'_z\cup S_x$ is of order f_z, $S_u=S'_u\cup S_x$ is of order f_u. Let $A_{ux}=(S_{yz}-S'_u)\cup S_y\cup S_z$, $A_{zx}=(S_{yu}-S'_z)\cup S_y\cup S_u$, $A_{yx}=(S_{zu}-S'_y)\cup S_z\cup S_u$. We can calculate that $|A_{ux}|=c'$, $|A_{zx}|=b'$, $|A_{yx}|=a'$.

To complete the proof, let x be any vertex in the complete $PC(2k)$-distance graph. Let y be the vertex which is obtained from x by adding all the elements of A_{xy} to x. Similarly define z and u. It can now be readily checked that the subgraph induced by x, y, z and u is isomorphic to the (K_4, w) we started with. □

In support of Conjecture 5.2 we have the following theorem which is immediate corollary of Theorem 4.3 and Theorem 5.8.

Theorem 5.9. *The projective cube of dimension $2k$, $PC(2k)$, is a graph of odd-girth $2k+1$ which bounds the class of partial 3-trees of odd-girth at least $2k+1$.*

Next we claim that $PC(2k)$ is a smallest graph of odd-girth $2k+1$ which may satisfy the statement of the previous theorem. This claim is implicitly proved in [18] based on the following notations.

Definition 5.10. Given a graph G of odd-girth $2k+1$ and an integer l, $l<k$, the walk power $G^{(2l+1)}$ of G is defined to be a graph on $V(G)$ where vertices u and v are adjacent if there is a $(u-v)$-walk of length $2l+1$.

Given an odd number $2l+1$, traversing an edge uv an odd number of times is a $(u-v)$-walk of length $2l+1$. Thus G is a subgraph of $G^{(2l+1)}$ for any choice of l. This is one of the main reasons we only considered walks of odd-length in this definition. The other reason is that the choice of odd number $2l+1$ and the condition of G being of odd-girth at least $2l+3$ insure that G has no loops. The following easy lemma is the main interest of this definition for us:

Lemma 5.11. *Let G and H be two graphs of odd-girth at least $2k+1$ and let $\phi: V(G)\to V(H)$ be a homomorphism of G to H. Then ϕ is also a homomorphism of $G^{(2k-1)}$ to $H^{(2k-1)}$.*

Thus if a graph B of odd-girth $2k+1$ bounds a class $\mathcal{C}$ of graphs, then $|V(B)|$ is an upper bound on the order of clique numbers of graphs in $\{G^{(2k-1)}|G \in \mathcal{C}\}$. To prove that $PC(2k)$ has the smallest order among graphs of odd-girth $2k+1$ bounding $\mathcal{PT}_{3,2k+1}$ we prove that this class of graphs has some weak sense of perfectness. That is to say, there is a graph $G \in \mathcal{PT}_{3,2k+1}$ for which the clique number of $G^{(2k-1)}$ is 2^{2k}. This approach was introduced in [18] where a planar graph G of odd-girth $2k+1$ satisfying $w(G^{(2k-1)}) = 2^{2k}$ was built. Our observation here is that the construction given there is also a partial 3-tree. Thus we only describe the construction to verify that the result is a partial 3-tree, and refer the reader to [18] for the verification of $w(G^{(2k-1)}) = 2^{2k}$.

Theorem 5.12. *There exists a partial 3-tree G of odd-girth $2k+1$ for which the graph $G^{(2k-1)}$ has clique number 2^{2k}.*

Sketch proof: We consider the following construction of a planar graph G of odd-girth $2k+1$ whose walk power $2k-1$ has a clique of order 2^{2k}. The graph G is built iteratively starting with a particular subdivision G_0 of K_4 which, when drawn on the plane, all four faces are odd-cycles of order $2k+1$. It is shown that for any such subdivision, the walk power $2k-1$ is a complete graph on $4k+2$ vertices. Given already constructed G_i, if there is a $u-v$ path P all whose internal vertices are of degree 2, then first a new $u-v$ path P' of the same length as P and parallel to it is added so that together with P they make a new face of the plane graph, and then an internal vertex of P' is connected to a particular internal vertex of P such that the PP' face is split into two faces each being a $(2k+1)$-cycle, thus having graph G_{i+1}. If $G_i^{(2k-1)}$ has a clique that uses l internal vertices of P, then we will find an extension of it in $G_{i+1}^{(2k-1)}$ which uses l more vertices. With a specific choice of a subdivision of K_4 and a specific rule on connecting P and P', at the end of this process we will have a graph G for which $G^{(2k-1)}$ has a 2^{2k}-clique.

What we observe furthermore are a few easy facts whose proofs we leave as exercise:

* K_4 is the first 3-tree.
* If G is a partial 3-tree and e an edge of it, any subdivision of e also implies a partial 3-tree.
* If G is a partial 3-tree and P is a path whose internal vertices are of degree 2, then adding a vertex which is joined to (only) three internal vertices of P is also a partial 3-tree.

One can view the construction of [18], sketched above, as repeated applications of these three steps. Thus the resulting graph at each step, and in particular at the final step, beside being a planar graph is also a partial 3-tree. Thus we have constructed a partial 3-tree G of odd-girth $2k+1$ for which $G^{(2k-1)}$ has clique number at least 2^{2k}. That the clique number of $G^{(2k-1)}$ is indeed 2^{2k} and not bigger follows from the fact that G maps to $PC(2k)$ by Theorem 5.9 (this fact was not verified in [18]).

In summary we have proved:

Theorem 5.13. *The projective cube of dimension $2k$, $PC(2k)$, is a smallest graph, in term of the number of vertices, of odd-girth $2k+1$ which bounds the class $\mathcal{PT}_{3,2k+1}$.*

6. Application to edge-coloring

The edge-chromatic number of a multigraph G, denoted $\chi'(G)$ is the smallest number of matchings to which $E(G)$ can be partitioned to. The fractional edge-chromatic number of G, denoted $\chi'_f(G)$, is the smallest total sum of weights assigned to the matchings of G such that total weight of matchings containing each given vertex is at least one. Given a subset S of odd number of vertices, it can be verified that $\frac{2|E(G[S])|}{|S|-1}$ is a lower bound for $\chi'_f(G)$. Let $\Lambda(G) = \max_{|S| odd}\{\frac{2|E(G[S])|}{|S|-1}\}$. We refer to text book [21] for more fractional coloring and a proof and references to the following theorem:

Theorem 6.1. *Given a multigraph G we have $\chi'_f(G) = \max\{\Delta(G), \Lambda(G)\}$.*

Using this theorem it can be easily verified that a cubic multigraph has fractional edge-chromatic number exactly 3 if and only if it is bridgeless. Thus Tait's classic reformulation of the four-color theorem ([25], then a conjecture) is to say that edge-chromatic number of cubic bridgeless planar multigraph is equal to its fractional edge-chromatic number. With such a view P. Seymour proposed the following strong generalization:

Conjecture 6.2. *[22] For every planar multigraph we have* $\chi'(G) = \lceil \chi'_f(G) \rceil$.

The conjecture is proved for K_4-minor-free graphs in [24] ([9] provide a different proof) and using this result it is proved in [12] for the larger class of K_5^--minor-free graphs where K_5^- is the graph obtained from K_5 by removing one edge. As a direct generalization of Tait's statement, the case of k-regular planar multigraphs satisfying $\chi'_f(G) = k$ has got more attention. After verifying condition under which a k-regular multigraph satisfies $\chi'_f(G) = k$, this special case of the conjecture can be restated as:

Conjecture 6.3. *If G is a planar k-regular multigraph in which any set of X of odd number of vertices is connected to $V(G) \setminus X$ by at least k edges, then G is k-edge-colorable.*

The cases $k = 4, 5$ of this conjecture are proved in [10], the case $k = 6$ in [7], the case $k = 7$ in [8] (see also [4]) and the case $k = 8$ in [5]. Proofs are based on induction on k, thus dependent on the case of $k = 3$ which is equivalent to the four-color theorem. The case $k = 4$ is known to imply the four-color theorem and one cannot expect an independent proof, however it is not known if the cases $k \geq 5$ are stronger than the four-color theorem, though we expect them to be.

In this section, providing support for the general case of Conjecture 6.3, and as an application of our work, we prove that Conjecture 6.3 holds on the subclass of $(2k + 1)$-regular planar multigraphs whose duals are partial 3-trees. This subclass of planar graphs can be characterized as a minor closed family of graphs with four forbidden minors given in Figure 4. Observe that two of these forbidden minors are K_5 and $K_{3,3}$ whose absence as a minor implies planarity and the other two are duals of the two planar graphs of Figure 1.

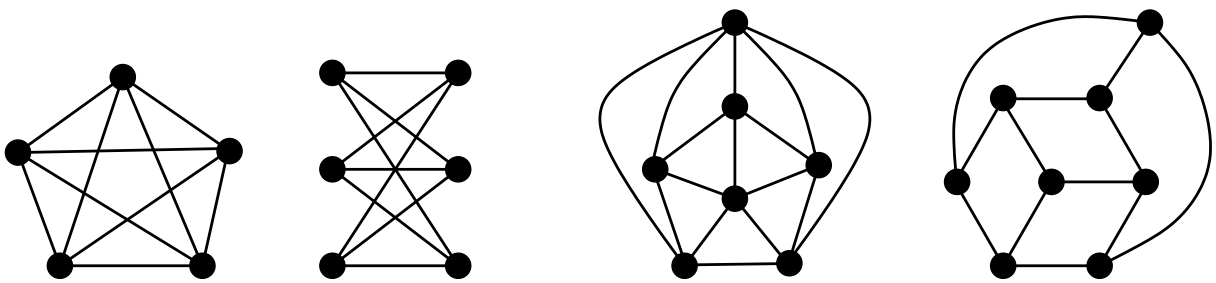

Figure 4: Forbidden minors of the class planar graphs whose duals are partial 3-trees

Theorem 6.4. *Let G be a planar $(2k+1)$-regular multigraph whose dual is a partial 3-tree. Furthermore, assume for each subset X of odd number of vertices the number of edges connecting X to $V(G) \setminus X$ is at least $2k + 1$. Then $\chi'(G) = 2k + 1$.*

Proof. Let G be a planar $(2k + 1)$-regular multigraph together with a planar embedding and let G^D be its duals with corresponding planar drawing. Let X be a set of vertices of G and let $\delta(X)$ be the set of edges with one end in X and another end in $V(G) \setminus X$. Each vertex of G corresponds to a facial cycle of G^D which is a cycle of length $2k + 1$. More generally given a set X of vertices of G, $\delta(X)$ corresponds to a cycle C_X of length $|\delta(X)|$ which bounds $|X|$ number of faces of G^D. As all of the facial cycles are of odd-length, the parity of C_X is determined by the parity of $|X|$. In other words, C_X is an odd-cycle if and only if X contains an odd-number of the vertices of G. Thus the condition on G that for every set X of odd number of vertices $\delta(X)$ has at least $2k + 1$ edges is equivalent to the statement that G^D has odd-girth $2k + 1$.

Since G^D has odd-girth $2k + 1$, and since we have assumed it is a partial 3-tree, by Theorem 5.9 it admits a homomorphism to $PC(2k)$. In any such mapping the image of any $(2k + 1)$-cycle of G^D is a $(2k + 1)$-cycle of $PC(2k)$. Recall that edges of $PC(2k)$ are colored by the elements of the $(2k + 1)$ set $S_{2k} = \{e_1, e_2, \ldots, e_{2k}, J\}$ where each $(2k + 1)$-cycle receives each color exactly once. Let ϕ be a mapping of G^D to $PC(2k)$ and color edges of G^D using S_{2k} as the color set, assigning to edge e the color of $\phi(e)$ in $PC(2k)$. Consider a facial cycle C_v corresponding to a vertex v of G. As v is of degree $2k + 1$, C_v is a $(2k + 1)$-cycle which maps to a $(2k + 1)$-cycle of $PC(2k)$. By Theorem 5.3 the set of colors assigned to C_v is the full set S_{2k} having each color used exactly once. We may now color edges of G by assigning to each edge the color of its corresponding edge in G^D. As edges incident to a vertex v corresponds to edges of a $(2k + 1)$-cycle of G^D, they receive distinct colors. □

We restate the theorem also using the set of four forbidden minors:

Theorem 6.5. *Let G be a $(2k+1)$-regular multigraph which does not contain any of the four graphs of Figure 4 as a minor. Furthermore assume for each subset X of odd number of vertices the number of edges connecting X to $V(G) \setminus X$ is at least $2k+1$. Then $\chi'(G) = 2k+1$.*

7. Remark and discussions

At the end we have a few remarks:

1. We gave an independent proof that $PC(2k)$ is triple-transitive. The proof was based on the fact that given triples x, y, z and subsets A, B and C of S_{2k} corresponding to three edges of the triangle induced by xyz in $PC(2k)$-distance complete graph, if $|A| + |B| + |C|$ is odd, then each element of S_{2k} is either in all three of them or exactly in one of them. Furthermore, the number of elements in all three of them is only a function of distances among x, y and z. But if a set of four vertices are selected, then common elements among six sets corresponding to six edges of the K_4 is no longer determined uniquely by the distances of the six pairs. Thus we can easily show that $PC(2k)$ is not 4-tuple-transitive.

2. We find it rather surprising that a sort of perfectness holds for the class of partial 3-trees in the following sense. The order of a smallest graph of odd-girth $2k+1$ to which every partial 3-tree of odd-girth at least $2k+1$ admits a homomorphism, the largest chromatic number of $G^{(2k-1)}$ when G is a partial 3-tree of odd-girth at least $2k+1$ and the largest clique number of $G^{(2k-1)}$ when G is a partial 3-tree of odd-girth at least $2k+1$ are all 2^{2k}. This is not necessarily the case for the class of partial t-trees, $t \neq 3$. For example for the class partial 2-trees, which is the same as the class of K_4-minor-free graphs, the smallest triangle-free graph to which every triangle-free graph with no K_4-minor admits a homomorphism is of order 8, but the largest clique one can find in a $G^{(3)}$ where G is a triangle-free graph with no K_4-minor is 6 (ongoing work of W. He, second author and Q. Sun, [11]).

3. We showed that $PC(2k)$ has the smallest number of vertices among all graphs of odd-girth $2k+1$ which bounds all partial 3-trees of odd-girth at least $2k+1$. It follows from a simple construction in [15] that any such a minimal bound must be of minimum degree at least $2k+1$. Thus $PC(2k)$ is also optimal in term of the number of edges.

4. Using terminology and discussions of [16] and [17], the question of bounding signed bipartite partial t-trees will be addressed in a forthcoming work. Using such results we will be able to present $2k$-regular analogue of Theorem 6.4.

5. We would like to point out that, using Theorem 4.3, given fixed values of k and t one can check in a polynomial time if an input graph B of odd-girth $2k+1$ bounds the class of all partial t-trees of odd-girth at least $2k+1$. Such algorithmic applications of this work will be discussed in forthcoming works.

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